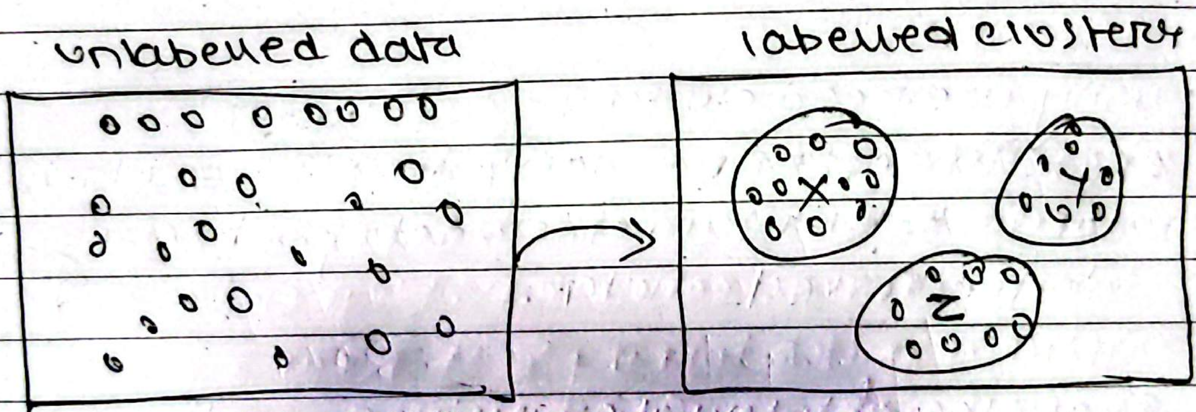
Unsupervised Model Evaluation

Unsupervised model evaluation: it is the process of checking how well an ~~so~~ unsupervised learning algorithm discover patterns, groups or relationships in data with predifined target labels.

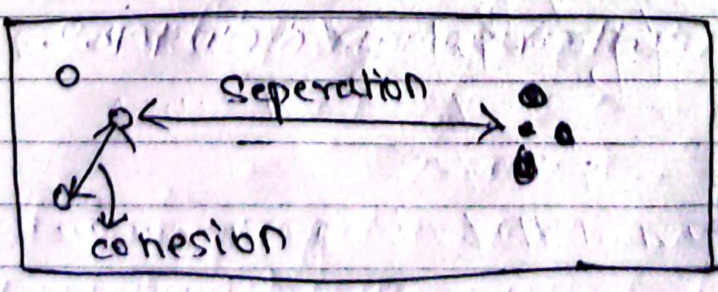
Algorithm for ~~sup~~ unsupervised model Evation:

1) Clustering: Grouping the large unlabelled data into smaller groups based on similarity of data.



Cohession: How close data points are to each other inside the same cluster.

Seperation: How far the different clusters are from each other.



Note: A good cluster should have high cohession and high seperation.

Q. How do we know whether our cluster are actually good?

Ans:- By using clustering evaluation.

* Types of clustering Evaluation:

1) Internal clustering Evaluation: It is used when true labels are not available. It evaluates the quality of clustering using the data itself.

Q. Why we need clustering Evaluation?
Ans:- Since, we usually don't have actual labels to tell us whether data point belongs to the correct cluster.

Examples of Internal clustering evaluation are:

i) Silhoutte Score: It is used to measure both cohession and seperation.

Q. Why we need Silhoutte Score?
Ans:- to measure how well each data points fits into their assigned clusters.

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Numerical Example

For a data point,

a = average distance to points in its own cluster

b = average distance to the nearest other cluster.

Formula:

Silhouette = (b - a) / Max(a, b)

Eg:

a = 2

b = 6

$\therefore$ S = (6 - 2) / 6

S = 0.67

Interpretation

Closer to +1 → Good clustering

Around 0 → Overlapping clusters

Negative → possibly wrong cluster

Since, 0.67 is greater than 0 and closer to +1

so, it is good clustering.

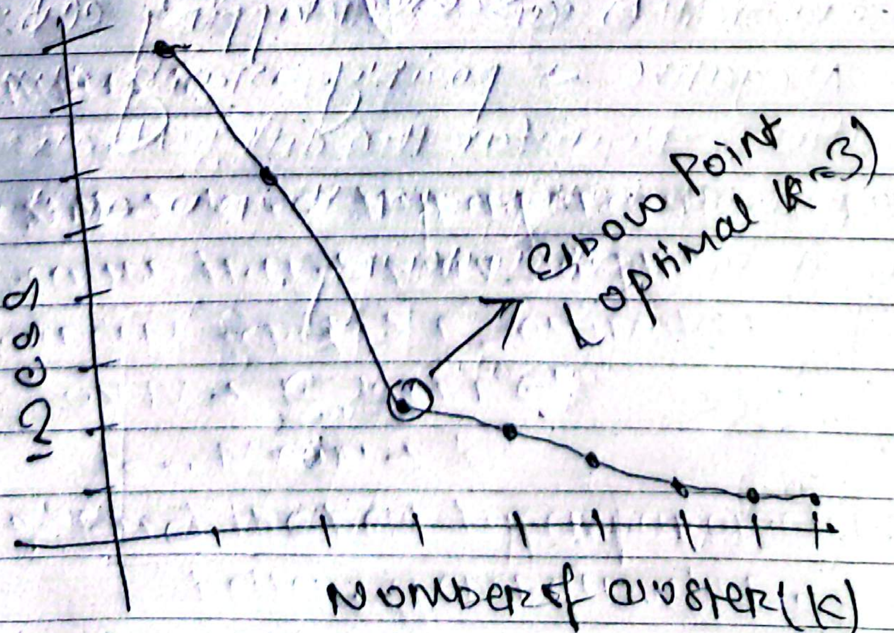
Other Internal Metrics

2) Davies-Bouldin Index (DBI) : Average similarity betwn each cluster and it's most similar other cluster. lower is better (0 is best possible score)

3) Elbow Method (using WCSS) : run clustering for k = 1, 2, 3 .... and plot WCSS. pick the k where the curve bends like an elbow, after which adding more clusters barely helps.

4) Calinski-Harabasz Index : ratio of betwn cluster seperation to within cluster cohesion. Higher is better

Elbow's Graph



2) External clustering Evaluation : External evaluation compares the generated cluster with known / true class labels.

Examples

1) Adjusted Rand index.
2) Normalized Mutual Information
3) Purity